NOTE If you're familiar with SOAP from the old ASP.NET days, you can think of OpenAPI as being the HTTP/REST equivalent of Web Service Description Language (WSDL). Just as a .wsdl file described your XML SOAP services, so the OpenAPI document describes your REST API.

ASP.NET Core includes some support for OpenAPI documents out of the box, but to take advantage of them you'll need to use a third-party library. The two best-known libraries to use are called NSwag and Swashbuckle. In this chapter I use Swashbuckle to add an OpenAPI document to an ASP.NET Core app. You can read how to use NSwag instead at http://mng.bz/6Dmy.

NOTE NSwag and Swashbuckle provide similar functionality for generating OpenAPI documents, though you'll find slight differences in how to use them and in the features they support. NSwag also supports client generation, as you'll see in section 11.4.

Add the Swashbuckle.AspNetCore NuGet package to your project by using the NuGet Package Manager in Visual Studio, or use the .NET CLI by running

```
dotnet add package Swashbuckle.AspNetCore
```

from your project's folder. Swashbuckle uses ASP.NET Core metadata services to retrieve information about all the endpoints in your application and to generate an OpenAPI document. Then this document is served by middleware provided by Swashbuckle, as shown in figure 11.1. Swashbuckle also includes middleware for visualizing your OpenAPI document, as you'll see in section 11.2.